I. Physics

Measurement/Unit

Notes

- A unit of measurement is a definite magnitude of a quantity, defined and adopted by convention or by law.
- It is used as a standard for measurement of the same kind of quantity.
- Any other quantity of that kind can be expressed as a multiple of the unit of measurement.
- To measure physical quantities four systems are in application as follows :

1. CGS System (Centimetre - Gram - Second System) :

- It is a variant of the metric system based on the

Centimetre	–	unit of length
Gram	–	unit of mass
Second	–	unit of time

2. FPS System (Foot - Pound - Second system) :

- It is also known as the British System of measurement.
- It relates to measuring –

Foot	–	unit of length
Pound	–	unit of mass
Second	–	unit of time

Note : 1 Pound is equal to 453.59 grams.

3. MKS system (Metre - Kilogram - Second system) :

- It relates to measuring –

Metre	–	unit of length
Kilogram	–	unit of mass
Second	–	unit of time

4. SI system (International System of Units) :

- It is the modern form of the metric system of measurement.
- It is the most widely used system of measurement.
- It was adopted during the conference on International Weight and Measures, held at Paris (France) in 1960.
- It has seven fundamental or base units.

Details of Fundamental units :

S.No.	Physical Quantity	S.I. Unit	Abbreviation
1.	Length	Metre	m
2.	Mass	Kilogram	kg
3.	Time	Second	s
4.	Temperature	Kelvin	K
5.	Luminous Intensity	Candela	cd
6.	Electric Current	Ampere	A
7.	Amount of Substance	Mole	mol

Definition of Units of SI System :

1. Metre (m) :

- The path travelled by light in vacuum during a time interval of 1/299,792,458 second. It is defined by taking the fixed numerical value of the speed of light in vacuum '*c*' to be 299,792,458 m/s.

2. Kilogram (kg) :

- As per old definition, one kg is the mass of Platinum - Iridium prototype kept at Sevres (Paris). Generally, at 4ºC, the mass of 1 litre pure water is 1 kilogram.

According to revised definition (which came into force on 20th May, 2019), **the kilogram** is now defined in terms of the Planck constant. It is defined by taking the fixed numerical value of the Planck constant '*h*' to be $6.62607015 \times 10^{-34}$ kg m^2/s. Thus,

$$1 \text{ kg} = \left(\frac{h}{6.62607015 \times 10^{-34}}\right) m^{-2} s$$

3. Second (s) :

- One second equals the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the unperturbed ground state of Caesium (Cs-133) atom. It is defined by taking the fixed numerical value of the caesium frequency to be 9,192,631,770 Hz, which is equal to s^{-1}.

4. Ampere :

- As per old definition; given two parallel, rectilinear conductors of negligible circular cross-section positioned 1 metre apart in vacuum, one ampere is the electric current which passes through both of them makes them attract each other by the force of 2×10^{-7} Newton per metre of length.

According to revised definition (which came into force on 20th May, 2019), **the ampere** is defined by taking the fixed numerical value of the elementary charge 'e' to be 1.602176634 × 10^{-19} As (ampere second). Thus, one ampere is the electric current corresponding to the flow of 1/(1.602176634 × 10^{-19}) elementary charges per second.

5. Kelvin :

- As per old definition, one kelvin equals 1/273.16 of the thermodynamic temperature of the triple point of water.

According to revised definition (which came into force on 20th May, 2019), **the kelvin** is defined by taking the fixed numerical value of the Boltzmann constant '*k*' to be 1.380649 × 10^{-23} kg $m^2s^{-2}K^{-1}$. Thus,

$$1\text{ K} = \left(\frac{1.380649}{k}\right)\times 10^{-23}\ kg\ m^2 s^{-2}$$

That means one kelvin is equal to the change of thermodynamic temperature that results in a change of thermal energy *k*T by 1.380649 × 10^{-23} Joule (kgm^2s^{-2}).

6. Candela (cd) :

- The candela is the luminous intensity in a given direction of a source that emits monochromatic radiation of frequency 540 × 10^{12} hertz and has a radiant intensity in that direction of 1/683 w/sr.

7. Amount of Substance - Mole (mol) :

- As per old definition, one mole is the amount of substance composed of as many specified elementary units (molecules)/atoms as there are atoms in 0.012 kg of C-12.

According to revised definition (which came into force on 20th May, 2019), **the mole** is the amount of substance of a system that contains 6.02214076 × 10^{23} specified elementary entities. This number is the fixed numerical value of the Avogadro constant 'N_A', when expressed in the unit mol^{-1} and is called the Avogadro number.

Derived units :

- These units of measurement are derived from the 7 base units, specified by the International System of Units.
- These are either dimensionless or can be expressed as a product of one or more of the base unit, possibly scaled by an appropriate power of exponentiation.
- Some of the coherent derived units in the SI are given special names. Below table lists 22 SI units with special names. Together with the seven base units, they form the core of the set of SI units. All other SI units are combinations of some of these 29 units.

The 22 SI derived units with special names and symbols

Derived quantity	Special name of unit	Unit expressed in terms of base units	Unit expressed in terms of other SI units
plane angle	radian	rad = m/m	-
solid angle	steradian	sr = m^2/m^2	-
frequency	hertz	Hz = s^{-1}	-
force	newton	N = kg m s^{-2}	-
pressure, stress	pascal	Pa = kg m^{-1} s^{-2}	-
energy, work, amount of heat	joule	J = kg m^2 s^{-2}	N m
power, radiant flux	watt	W = kg m^2 s^{-3}	J/s
electric charge	coulomb	C = A s	-
voltage/electric potential difference	volt	V = kg m^2 s^{-3} A^{-1}	W/A
capacitance	farad	F = kg^{-1} m^{-2} s^4 A^2	C/V
electric resistance	ohm	W= kg m^2 s^{-3} A^{-2}	V/A
electric conductance	siemens	S = kg^{-1} m^{-2} s^3 A^2	A/V
magnetic flux	weber	Wb = kg m^2 s^{-2} A^{-1}	V s

magnetic flux density	tesla	$T = kg\ s^{-2}\ A^{-1}$	Wb/m^2
inductance	henry	$H = kg\ m^2\ s^{-2}\ A^{-2}$	Wb/A
Celsius temperature	degree Celsius	ºC = K	-
luminous flux	lumen	lm = cd sr	cd sr
illuminance	lux	$lx = cd\ sr\ m^{-2}$	lm/m^2
activity referred to a radionuclide	becquerel	$Bq = s^{-1}$	-
absorbed dose, kerma	gray	$Gy = m^2\ s^{-2}$	J/kg
dose equivalent	sievert	$Sv = m^2\ s^{-2}$	J/kg
catalytic activity	katal	$kat = mol\ s^{-1}$	-

Some derived units in the SI expressed in terms of base units

Derived quantity	Derived unit expressed in terms of base units
area	m^2
volume	m^3
speed, velocity	$m\ s^{-1}$
acceleration	$m\ s^{-2}$
wavenumber	m^{-1}
density, mass density	$kg\ m^{-3}$
surface density	$kg\ m^{-2}$
specific volume	$m^3\ kg^{-1}$
current density	$A\ m^{-2}$
magnetic field strength	$A\ m^{-1}$
amount of substance concentration	$mol\ m^{-3}$
mass concentration	$kg\ m^{-3}$
luminance	$cd\ m^{-2}$

Other measures of Length unit :

i. 1 milimetre = 10^{-3} metre

ii. 1 centimetre = 10^{-2} metre

iii. 1 kilometre = 1000 metre

iv. 1 Angstrom (Å) = 10^{-10} metre

- A unit of length to measure very small distances like wavelength, atomic and ionic radius or size of molecules and spacing between planes of atoms in crystals.

v . Nanometre –

- 1 nanometre is one billionth of a metre, equal to 10^{-9} metre.
- It is used to measure extremely small objects such as atomic structures or transistors found in modern CPUs.

vi . Micron –

- It is the previous name of micrometre.
- 1 micron is equal to 10^{-6} m.
- It is represented by (μ).
- It is used to measure cell size.

vii . Astronomical Unit –

- Astronomical unit is usually used to measure distances within our solar system.
- An astronomical unit (AU) is the average distance between Earth and Sun.
- It equals to approximately 150 million kilometre or 1 AU = 1.496×10^{11} metre.

viii . Light-year –

- A light-year is a distance that light travels in vacuum in one Julian year.
- It is a unit of length used to express astronomical distances.
- 1 light-year is equal to
  $= 3 \times 10^8$ m/s $\times 365.25 \times 24 \times 60 \times 60$s
  $= 9.461 \times 10^{15}$ metre.
- Julian year : In Astronomy, a Julian year is a unit of measurement of time defined as exactly 365.25 days of 86400 SI seconds each.

ix. Parsec –

- A Parsec is an astronomical term used to measure large distances to astronomical objects outside the solar system.
- A Parsec is defined as the distance at which one astronomical unit subtends an angle of one arcsecond, which corresponds to $648000/\pi$ astronomical units.
- 1 Parsec is equal to 3.0857×10^{16} metre, or
- 1 Parsec is equal to 3.262 light-years.

Question Bank

1. In the following which is fundamental physical quantity?

(a) Force (b) Velocity
(c) Electric current (d) Work
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

In physics, there are seven fundamental physical quantities (which are measured in base or fundamental units) : length, mass, time, electric current, temperature, luminous intensity and amount of a substance.

2. The unit of power is –

(a) Hertz (b) Volts
(c) Watt (d) Neutron

U.P.P.S.C. (GIC) 2010

Ans. (c)

In physics, power is the rate of doing work or of transferring heat or electrical energy i.e. the amount of energy transferred or converted per unit time.

$$\text{Power (P)} = \frac{\text{Work (w)}}{\text{Time (t)}}$$

The SI unit of power is watt, which is equal to joule per second. Power is always represented in watt (W) or Kilowatt (KW).

3. The unit of electric power is :

(a) Ampere (b) Volt
(c) Coulomb (d) Watt
(e) None of the above / More than one of the above

65th B.P.S.C. (Pre) 2019
64th B.P.S. (Pre) 2018

Ans. (d)

Electric power is the rate, per unit time, at which electrical energy is transferred by an electric circuit. The SI unit of electric power is watt. Ampere is the unit of electric current, volt is the unit of electrical potential and coulomb is the unit of electric charge in the SI system.

4. The unit of the force is –

(a) Faraday (b) Fermi
(c) Newton (d) Rutherford

M.P.P.C.S. (Pre) 1990

Ans. (c)

The SI unit of Force is 'Newton' or kg.m/sec^2.
Force = mass × acceleration
In physics, something that causes a change in the motion of an object is called force. The modern definition of force (an object's mass multiplied by it's acceleration) was given by Isaac Newton in his laws of motion.

5. The unit of work is :

(a) Joule (b) Neutron
(c) Watt (d) Dyne

U.P.P.C.S. (Pre) 1996

Ans. (a)

When a force acts to move an object, then work done by the force is equivalent to the product of force and displacement in the direction of force . It is a scalar quantity. The SI unit of work is Newton metre, which is also called as joule. Joule is also the unit of Energy.

6. Frequency is measured in :

(a) hertz (b) metre/second
(c) radian (d) watt
(e) None of the above / More than one of the above

64th B.P. S.C. (Pre) 2018

Ans. (a)

The Hertz (symbol : Hz) is the derived unit of frequency in the International System of Units (SI system) and is defined as one cycle per second.

7. What is measured in hertz?

(a) Frequency (b) Energy
(c) Heat (d) Quality
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

See the explanation of above question.

8. The term GHz is the indicator of which feature of the computer?

(a) Number of Pixels (b) Resolution
(c) Speed (d) Storage

Uttarakhand P.C.S. (Pre) 2024

Ans. (c)

GHz stands for gigahertz, which is a unit of frequency in the International System of Units (SI). It's equal to one billion hertz and is used to measure how quickly an electronic device operates. When it comes to computers, GHz measures the speed at which electronic components process instructions from software. The term GHz (gigahertz) primarily indicates the clock speed or processing speed of a computer's central processing unit (CPU) i.e. the number of cycles CPU executes per second.

9. The SI unit of electrical resistivity of conductor is –

(a) Faraday (b) Volts
(c) Ampere (d) Ohm

M.P.P.C.S. (Pre) 1993

Ans. (*)

Electrical resistivity is an intrinsic property that quantifies how strongly a given material opposes the flow of electric current. A low resistivity indicates a material, that readily allows the movement of electric charge. The SI unit of electrical resistivity is ohm-meter (Ωm). It is commonly represented by the – Greek letter ρ (rho) defined as $\left[\rho = \frac{RA}{l}\right]$

Here,
R = electrical resistance of the material
l = Length, A = Cross section area, ρ = resistivity
As per other options, Faraday is the SI unit of Capacitance, Volt is the SI unit of Electric Potential, Ampere is the SI unit of Electric Current while Ohm is the SI unit of Electrical Resistance (not resistivity).

10. 'Ohm-meter' is unit of :
(a) Resistance (b) Conductance
(c) Resistivity (d) Charge
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

See the explanation of above question.

11. What is the unit of measure of magnetic field?
(a) Cobalt (b) Ampere
(c) Ohm (d) Tesla

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

A magnetic field (sometimes called B-field) is a physical field that describes the magnetic influence on moving electric charges, electric currents, and magnetic materials.
In electromagnetics, the term magnetic field is used for two distinct but closely related vector fields denoted by the symbols **B** and **H**. In the International System of Units (SI), the unit of **B**, magnetic flux density (also called magnetic B-field), is the tesla (Symbol : T; in SI base units : kilogram per second squared per ampere), which is equivalent to newton per ampere-meter. The unit of **H**, magnetic field strength, is ampere per meter (A/m). B and H differ in how they take the medium and/or magnetization into account.

12. Which of the following is equivalent to tesla ?
(a) Ampere per Newton
(b) Newton per coulomb
(c) Newton per ampere-second
(d) Newton per ampere-meter

70th B.P.S.C. (Pre) 2024

Ans. (d)

The tesla (symbol : T) is the standard unit of magnetic flux density (also known as magnetic induction or magnetic B-field) in the International System of Units (SI). One tesla is equal to one weber per square meter (1 Wb/m^2) or one Newton per ampere-meter (1N/amp.m). The unit is named after Nikola Tesla, a Serbian-American inventor and electrical engineer.

13. Which of the following pairs is NOT correctly matched?

Quantity		**S.I. Unit**
(a) Power of lense	–	Diopter
(b) Pressure	–	Pascal
(c) Activity of radio-active substance	–	Curie
(d) Heat	–	Joule

U.P. P.C.S. (Pre) 2022

Ans. (c)

Unit of radioactivity is defined as the activity of a quantity of radioactive material where one decay takes place per second. The SI unit of radioactivity is becquerel (Bq) and this term is named after Henri Becquerel. Curie (Ci) is the old U.S. unit (non-S.I. unit) of radioactivity. Other pairs are correctly matched.

14. Light-year is the unit of –
(a) Distance (b) Time
(c) Speed of light (d) Intensity of light

U.P. U.D.A./L.D.A. (Pre) 2013
R.A.S./R.T.S. (Pre) 1997
M.P.P.C.S. (Pre) 2008
Jharkhand P.C.S. (Pre) 2013, 2023
66th B.P.S.C. (Pre) (Re. Exam) 2020
66th B.P.S.C. (Pre) 2020

Ans. (a)

Light-year is a unit of astronomical distance equivalent to the distance that light travels in one year, which is 9.46×10^{15} metres.

15. A light-year is the –
(a) Year which had maximum sunlight
(b) Year in which workload was very light
(c) Distance travelled by light in one year
(d) Mean distance between Sun and Earth

U.P. Lower Sub. (Mains) 2013

Ans. (c)

See the explanation of above question.

16. 'Light-Year' is –
(a) The year in which February has 29 days
(b) The distance travelled by light in one year
(c) The time which sun rays take to reach the earth
(d) The time in which a spacecraft reaches moon from the earth

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (b)

See the explanation of above question.

17. Which one of the following is a reason why astronomical distances are measured in light-years?
(a) Distances among stellar bodies do not change
(b) Gravity of stellar bodies does not change
(c) Light always travels in straight line
(d) Speed of light is always same

I.A.S. (Pre) 2021

Ans. (d)

A light-year is the distance a beam of light travels in a vacuum in one Julian year. The speed of light is constant in a vacuum (not even affected by gravity). And that is where all the stellar bodies are present, in a vacuum. The speed of light is constant throughout the universe and is known to high precision. That is why all astronomical distances are measured in light-years. One light-year is equal to 9.461×10^{15} metre.

18. A parsec, a unit of distance used to measure the distance related to the stars in the sky, is equal to –

(a) 4.25 light-years (b) 3.25 light-years
(c) 4.50 light-years (d) 3.05 light-years

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

A parsec is a unit of length used to measure the astronomically large distances of objects beyond our solar system.
1 Parsec = 3.0857×10^{16} Metre
1 Light year = 9.461×10^{15} Metre
So, 1 Parsec = 3.262 light-years,
Now according to options, only (b) is close to the exact answer.

19. PARSEC is the unit of :

(a) Distance (b) Time
(c) Light intensity (d) Magnetic force

U.P.P.C.S. (Pre) 1997

Ans. (a)

See the explanation of above question.

20. Which unit of measurement is multiplied by 0.39 to convert it to 'inches' ?

(a) Millimetre (b) Centimetre
(c) Metre (d) Decimetre

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (b)

1 Centimetre = 0.39 inch, so we can multiply centimetre by 0.39 to convert it to 'inches'.

21. How can the height of a person who is six feet tall, be expressed (approximately) in nanometre?

(a) 183×10^6 nm (b) 234×10^6 nm
(c) 183×10^7 nm (d) 181×10^7 nm

I.A.S. (Pre) 2008

Ans. (c)

1 nanometre = 10^{-9} metre
1 Feet = 0.305 metre
1 Feet = 30.5×10^7 nanometre
6 Feet = $6 \times 30.5 \times 10^7$ nanometre = 183×10^7 nanometre

22. One nanometre is equal to –

(a) 10^{-6} cm (b) 10^{-7} cm
(c) 10^{-8} cm (d) 10^{-9} cm

U.P. U.D.A./L.D.A. (Pre) 2013
Uttarakhand P.C.S. (Pre) 2016

Ans. (b)

1 nanometre = 1.0×10^{-9} metre
Since, 1 metre = 100 cm
So, 10^{-9} metre = $10^2 \times 10^{-9}$ cm
= 10^{-7} cm

23. Ampere is the unit of –

(a) Voltage (b) Electric current
(c) Resistance (d) Power

Chhattisgarh P.C.S. (Pre) 2005

Ans. (b)

Ampere is the SI unit to measure electric current. If one ampere current is flowing in any conducting wire, it means 6.25×10^{18} electrons are entering per second from one direction and same amount of electrons flows from the other end per second.

24. Megawatt is the measuring unit of power which is –

(a) Generated (b) Consumed
(c) Saved (d) Lost in transmission

U.P. Lower Sub. (Pre) 1998

Ans. (a)

Megawatt is the measuring unit of power, which is generated in Power station or Power Plant. 1 Megawatt is equal to 10^6 (1 million) watt.

25. Match List-I with List-II and select the correct answer from the code given below :

List-I (Physical quantities)	**List-II (Units)**
A. Acceleration	**1. Joule**
B. Force	**2. Newton second**
C. Work done	**3. Newton**
D. Impulse	**4. Metre/second^2**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	1	2
(c)	2	3	4	1
(d)	4	3	1	2

U.P.P.C.S. (Pre) 2005
U.P. U.D.A./L.D.A. (Pre) 2001

Ans. (d)

Acceleration is the rate of change of velocity of an object, with respect to time. The SI unit of acceleration is metre per second square (m/s^2). The SI unit of force is newton (N). The SI unit of impulse is newton second (Ns). The SI unit of work is joule.

26. Which one of the following SI unit is not correctly matched?

(a) Work – Joule (b) Force – Newton
(c) Mass – kg. (d) Pressure – Dyne

U.P. Lower Sub. (Pre) 2013

Ans. (d)

Pascal is the unit of pressure or stress in the International System of Units (SI). Dyne is the unit of force in CGS system. Clearly, option (d) is not correctly matched.

27. Match List-I with List-II and select the correct answer using codes given below :

List-I (Units)	List-II (Parametric quantities)
A. Watt	**1. Heat**
B. Knot	**2. Navigation**
C. Nautical mile	**3. Speed of a ship**
D. Calorie	**4. Power**

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	1	2	3	4
(c)	4	3	2	1
(d)	2	4	1	3

U.P. U.D.A./L.D.A. (Pre) 2002

Ans. (c)

The SI unit of power is watt (W), which is equal to one joule per second. Knot is the unit of measuring of speed of a ship. Nautical mile is a unit of distance used by navigators in the sea. Calorie is a unit of measuring heat and energy. Thus option (c) correctly matched.

28. Match the following :

A. Joule	**1. Current**
B. Ampere	**2. Power**
C. Watt	**3. Work**
D. Volt	**4. Electric potential**

Code :

	A	B	C	D
(a)	3	1	2	4
(b)	1	2	3	4
(c)	4	3	2	1
(d)	1	3	2	4

U.P.P.C.S. (Pre) 1990

Ans. (a)

Joule is the SI unit of work. Ampere, watt and volt are the units of current, power and electric potential respectively.

29. How many watts are there in a horsepower?

(a) 1000 (b 750

(c) 746 (d) 748

M.P.P.C.S. (Pre) 1991

Ans. (c)

Horse power is a unit of measurement of power (the rate at which work is done).

1 watt = 1 joule/second

1 Horse power = 746 watt.

30. Which one of the following is not correctly matched ?

(a) Knot - Measure of speed of ship

(b) Nautical mile - Unit of distance used in navigation

(c) Angstrom - Unit of wavelength of light

(d) Light year - Unit of measuring time

U.P.P.C.S. (Mains) 2010

Ans. (d)

Light year is not a unit of measuring time, but a unit of measuring distance. The remaining pairs are correctly matched.

31. Angstrom is a unit of

(a) wavelength (b) energy

(c) frequency (d) velocity

(e) None of the above / More than one of the above

64th B.P.C.S. (Pre) 2018

Ans. (a)

Angstrom (Å) is the unit of length used mainly in measuring the wavelengths of electromagnetic waves which is equal to 10^{-10} metre or 0.1 nanometre. It is named after the 19th century Swedish physicist Anders Jonas Angstrom.

32. Match List-I (Quantity) with List-II (Units) and select the correct answer using the codes given below the lists:

List I	List II
A. High speed	**1. Mach**
B. Wavelength	**2. Angstrom**
C. Pressure	**3. Pascal**
D. Energy	**4. Joule**

Code :

	A	B	C	D
(a)	2	1	3	4
(b)	1	2	4	3
(c)	1	2	3	4
(d)	2	1	4	3

U.P.P.C.S. (Pre) 2006
I.A.S. (Pre) 1999

Ans. (c)

In fluid mechanics, the Mach number is a dimensionless quantity representing the ratio of the speed of a body to the speed of sound in the surrounding medium. Supersonic travell is a rate of an object that exceeds the speed of sound (Mach 1). So Mach is used to represent the high speed. Wavelength is measured in angstrom while pressure is measured in pascal and energy in joule.

33. 'Joule' is related to energy in the same way as 'Pascal' is related to :

(a) Mass (b) Pressure

(c) Density (d) Purity

(e) None of the above

Chhattisgarh P.C.S. (Pre) 2015

Ans. (b)

See the explanation of above question.

34. One micron is equal to –

(a) 1/10 mm (b) 1/100 mm

(c) 1/1000 mm (d) 1/10,000 mm

39th B.P.S.C. (Pre) 1994

Ans. (c)

1 micron $= 10^{-6}$ m.

$= 10^{-6} \times 10^{3}$ mm.

$= \frac{1}{10^3}$ mm.

$= \frac{1}{1000}$ mm.

35. One micron represents a length of –

(a) 10^{-6} cm (b) 10^{-4} cm

(c) 1 mm (d) 1 m

U.P.P.C.S. (Mains) 2011

Ans. (b)

One micron represents a length of 10^{-6} m. It is represented by μ sign.

1 micron = 0.000001 m. = 0.0001 cm.

$= \frac{1}{10000}$ cm. $= \frac{1}{10^4}$ cm. $= 10^{-4}$ cm.

36. Which one of the following is not correctly matched?

(a) Decibel – Unit of sound intensity

(b) Horsepower – Unit of power

(c) Nautical miles – Unit of naval distance

(d) Celsius – Unit of heat

U.P.P.C.S. (Spl.) (Mains) 2004

U.P.P.C.S.(Pre) 2001

Ans. (d)

Celsius is a scale and unit of measurement for temperature. Heat is measured in Calories. Nautical mile is used to measure distance at sea. 1 nautical mile is equal to 1.852 km. The SI unit of power is watt (W), which is equal to Joule per second. Another unit of power is horsepower (hp) or metric horsepower. 1 horsepower is equal to 746 watts. Decibel is used to measure the sound intensity.

37. Which one of the following is not the unit of heat?

(a) Calorie (b) Kilocalorie

(c) Kilojoule (d) Watt

M.P. P.C.S. (Pre) 2016

Ans. (d)

The watt is a derived unit of power in the International System of Units (SI). Rest of all are the units of heat.

38. Which one of the following is not the unit of heat?

(a) Centigrade (b) Calorie

(c) Erg (d) Joule

U.P. R.O./A.R.O. (Pre) 2017

Ans. (a)

Temperature is a physical quantity expressing hot or cold. It is measured with a thermometer calibrated in one or more temperature scales. The most commonly used scales are Centrigrade scale, Fahrenheit scale and Kelvin scale. While the calorie, erg and joule are the units of heat energy. Heat energy is transferred from one body to another as the result of a difference in temperature. Heat is a form of energy.

39. A distance of 1 km. means –

(a) 100 m. (b) 1000 cm.

(c) 1000 m. (d) 100 cm.

45th B.P.S.C. (Pre) 2001

Ans. (c)

The SI unit for distance is metre. 1 Km. is equal to 1000m. 1 m is equal to 100 cm.

40. One pikogram is equal to –

(a) 10^{-6} gram (b) 10^{-9} gram

(c) 10^{-12} gram (d) 10^{-15} gram

42nd B.P.S.C. (Pre) 1997

Ans. (c)

The smaller units of measuring mass are Milligram, Microgram, Pikogram and Femtogram.

1 Pikogram = 10^{-12} gram 1 Milligram = 10^{-3} gram

1 Microgram = 10^{-6} gram 1 Nanogram = 10^{-9} gram

1 Femtogram = 10^{-15} gram

41. Pascal is a unit of measuring :

(a) Humidity (b) Pressure

(c) Rainfall (d) Temperature

Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

The pascal (Pa) is the unit of pressure or stress in the International System of Units (SI). It is named after the scientist Blaise Pascal. One pascal is equivalent to one newton (N) of force applied over an area of one square metre ($1m^2$).

1 Pa = 1 newton/metre^2 = 1kg / ms^2 = 1kg $m^{-1}s^{-2}$

42. What is the unit of pressure?

(a) Newton / sq. metre (b) Newton-metre

(c) Newton (d) Newton/metre

(e) None of the above / More than one of the above

64th B.P.C.S. (Pre) 2018

Ans. (a)

See the explanation of above question.

43. **The unit of pressure is :**
(a) kg/cm^2 (b) kg/cm
(c) kg/mm (d) kg/cm^3
(e) None of the above/More than one of the above
65th B.P.S.C. (Pre) 2019

Ans. (e)

A kilogram-force per centimetre square (kgf/cm^2) is a deprecated unit of pressure which was often written as just kilogram per square centimetre (kg/cm^2). But, kg/cm^2 is not the correct expression for pressure and it should be kgf/cm^2. 1 kgf/cm^2 equals 98,066.5 pascals. Thus, in the present context, option (e) is the right answer.

44. **Which of the following are units of the pressure?**
A. bar **B. Pa**
C. torr **D. atm**
Choose the correct option :
(a) Only B and D (b) Only B, C and D
(c) Only A, B and D (d) All A, B, C and D
Rajasthan P.C.S. (Pre) 2024

Ans. (d)

The **pascal (Pa)** is the unit of pressure in the International System of Units (SI). It is defined as the pressure exerted by a force of 1 newton on a surface of 1 square meter (1 Pa = 1 N/m^2). The pascal is the reference unit for all other pressure units. Other commonly used pressure units are as follows :

- **Atmosphere (atm) :** Atmosphere is a unit of pressure used to measure atmospheric pressure. One atmosphere is equivalent to the pressure exerted by a 760 mm column of mercury at 0°C. 1 atm = 101,325 Pa.
- **The bar :** The bar is a unit of pressure commonly used to measure the pressure of tyres, oxygen and acetylene cylinders, as well as in scuba diving applications. One bar is equal to 0.9869 atmospheric pressure. 1 bar = 100,000 Pa.
- **The millimeter of mercury (mmHg) :** The millimeter of mercury, also known as the **torr**, is a unit of pressure historically used in physics. 760 mm Hg corresponds to one atmosphere at 0°C. 1 mm Hg = 133.322368 Pa at 0°C.
- **The psi (pound per square inch) :** The psi is a unit of pressure widely used to measure tyre pressure. 1 psi is equivalent to approximately 6894.76 Pa.

Hence, option (d) is the correct answer.

45. **What is the unit of atmospheric pressure?**
(a) Bar (b) Knot
(c) Joule (d) Ohm
Chhattisgarh P.C.S. (Pre) 2008

Ans. (a)

Bar is metric (but not SI) unit of pressure. It is equal to 10^5 newton/metre^2. Pascal is SI unit of pressure.
1 bar = 10^5 Pascal.

46. **1 kg/cm^2 pressure is equivalent to :**
(a) 0.1 bar (b) 1.0 bar
(c) 10.0 bar (d) 100.0 bar
Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

In this question 1 kg/cm^2 refers to 1 kg-force/cm^2.
1 dyne $= 1.02 \times 10^{-6}$ kg -force (kgf)
and 1 bar $= 10^6$ dyne/cm^2
Therefore substituting the value of dyne into the value of bar –
1 bar $= 10^6$ dyne/cm^2 $= 1.02 \times 10^{-6} \times 10^6$ Kgf/cm^2.
$= 1.02$ Kgf/cm^2.

47. **Which one of the following quantities does not have unit?**
(a) Stress (b) Force
(c) Strain (d) Pressure
(e) None of the above/More than one of the above
65th B.P.S.C. (Pre) 2019

Ans. (c)

When the shape or size of a matter is changed by applying an external force, it is known as strain. Since, it is a ratio, hence it is dimensionless and without any unit. The unit of force is newton (kg ms^{-2}) and the unit of stress and pressure is pascal (kg $m^{-1}s^{-2}$).

48. **1 barrel of oil is equals to which of the following?**
(a) 131 litre (b) 159 litre
(c) 179 litre (d) 201 litre
U.P.P.C.S. (Pre) 2009

Ans. (b)

The amount of oil is now measured in cubic metres.
1 barrel = 158.9873 litre
1 barrel = 0.158987 cubic metre
1 barrel = 42 U.S. gallon
1 barrel = 34.9723 U.K. gallons.

49. **The smallest unit of length is –**
(a) Micron (b) Nanometre
(c) Angstrom (d) Fermimetre
U.P.P.C.S. (Pre) 2005

Ans. (d)

1 Micron	$= 10^{-6}$ metre
1 Nanometre	$= 10^{-9}$ metre
1 Angstrom	$= 10^{-10}$ metre
1 Fermi (Femtometre)	$= 10^{-15}$ metre

50. **Match List-I with List-II and select the correct answer using the codes given below the lists.**

List - I	**List - II**
A. Cusec	**1. Pressure**
B. Byte	**2. Intensity of Earthquake**
C. Richter	**3. Rate of flow**
D. Bar	**4. Computer**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	2	1
(c)	4	3	2	1
(d)	3	4	1	2

U.P. Lower Sub. (Spl.) (Pre) 2008

Ans. (b)

Cusec is a measure of flow rate of water and is abbreviation of 'cubic feet per second' (28.317 litres per second). Byte is a unit of digital information in computing and communications that consists of eight bits. The Richter magnitude scale (also Richter scale) assigns a magnitude number to quantify the energy released by an earthquake. The bar is a metric (but not SI) unit of pressure exactly equal to 100,000 Pascal.

51. **What is measured in cusec ?**
(a) Purity of water (b) Depth of water
(c) Flow of water (d) Quantity of water

Uttarakhand P.C.S. (Pre) 2006

Ans. (c)

See the explanation of above question.

52. **NTU is the unit for measuring :**
(a) Pressure of water (b) Temperature of water
(c) Acidity of water (d) Turbidity of water

Rajasthan P.C.S. (Pre) 2023

Ans. (d)

Turbidity is described as the opaqueness of a fluid due to the presence of suspended solids and is measured in terms of nephelometric turbidity units (NTU). The higher the concentration of suspended solids in the water is, the dirtier it looks and the higher the turbidity is.

53. **Which of the following is the value of solar constant ?**
(a) 1.6 kW/m^2 (b) 1.4 kW/m^2
(c) 1.2 kW/m^2 (d) 1.8 kW/m^2

70th B.P.S.C. (Pre) 2024

Ans. (b)

The 'solar constant' is defined as the amount of solar energy received per unit surface at a distance of one astronomical unit (the average distance of Earth's orbit) from the Sun. The total solar irradiance (TSI) reaching Earth per unit area was originally thought to be a fixed value and was referred to as the solar constant. Over the years, measurement of this quantity by various techniques yielded values that were quite different from one another. Satellite TSI measurements have enabled significant convergence in scientific consensus on the TSI value, which observations from the currently operational NASA Solar Radiation and Climate Experiment (SORCE) have determined to be 1,361 watts per square meter (W/m^2). However, this value is not constant, but actually varies with time depending on solar activity, which is reflected in the counts of sunspots. It changes by ~0.1% in an 11-year solar cycle. Prior to the measurements obtained by the SORCE, the TSI value was estimated at 1366 W/m^2. The current TSI value from the TSIS-1 is 1361.6 ± 0.3 W/m^2 for the 2019 solar minimum. Hence, option (b) is the closest answer (1.4 kW/m^2 = 1400 W/m^2).

54. **Which one of the following is the unit of measure of the thickness of the ozone layer of the atmosphere ?**
(a) Knot (b) Dobson
(c) Poise (d) Maxwell

Uttarakhand P.C.S. (Pre) 2010

Ans. (b)

Ozone layer thickness (concentration) is expressed in terms of Dobson unit, which measure what its physical thickness would be if compressed in the Earth's atmosphere. 1 Dobson unit (DU) is defined to be 0.01 mm thickness at STP (Standard Temperature and Pressure). The unit is named after G.M.B. Dobson, one of the first scientists to investigate atmospheric ozone. One Dobson unit is equivalent to 2.687×10^{20} molecules of ozone per square metre.

55. **The thickness of the ozone in a column of air from the ground to the top of the atmosphere is measured in terms of :**
(a) Ozone unit (b) Thomson unit
(c) Dobson unit (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (c)

See the explanation of above question.

56. **'Dobson' Unit is used for the measurement of –**
(a) Thickness of Earth (b) Thickness of Diamond
(c) Thickness of Ozone layer (d) Measurement of Noise

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

See the explanation of above question.

57. **Which one among the following is measured by Dobson unit?**
(a) Ozone concentration (b) Thermal conductivity
(c) Soil moisture (d) Radiation

U.P. R.O./A.R.O. (Mains) 2021

Ans. (a)

See the explanation of above question.